

研究ミーティング

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研究目的

Microtearing乱流飽和レベルにおける磁場の圧縮効果を解明する.

- 非線形領域で圧縮性磁場のエネルギーがアルヴェニック磁場エネルギーとコンパラになる.
- MT乱流では電子温度揺動が卓越するため、磁場の圧縮性は無視できないはず（圧力場ラインスより）.
- 圧縮性磁場の効果を考慮したMT乱流飽和はこれまでに考えられていない（Low-beta オーダリングのため）.

研究方針

Drake+, 1980 に倣って シアフリー なシステム (self-filamentation mode) を採用する.

Drake 予測 スケーリング は MT 乱流 シミュレーション の 輸送 レベル を 再現 する (Doerk+, 2011). $B_{\perp}/B_0 \sim \rho_e/L_{T_{0e}}$

- $\phi, \delta B_{\parallel}$ を無視した SF モード の 輸送 レベル が Drake 予測 スケーリング に 合う か 確認.
- 磁場の圧縮効果を取り入れたスケーリング則の導出 と シミュレーション 結果 と の 比較.
-> MT 乱流 における 磁場の 圧縮効果 として 議論 する.

有限磁気シアの結果に対する課題

これまでの非線形計算において、圧縮磁場がMT乱流輸送レベルを抑制しうる結果が得られている。

以下の課題が積み残し。

- バースト現象.
- 既往研究のスケーリングとの比較.
- 複数コードによるクロスベンチマーク.
- 理論解析によるアプローチ.

⇒一旦保留

シミュレーションの道のり

- Self-filamentation不安定な波数領域を同定.
- $\phi, \delta B_{\parallel}$ 無視した場合の非線形計算を異なる温度勾配に対して実施.
- Drakeの輸送スケージングの比較
- 磁場の圧縮性を考慮した非線形計算の実施.



- ✓ Self-filamentation不安定な波数領域を同定.
- ✓ AstroGKとGS2で結果の比較.
- ✓ 非線形計算を実行中.

Self-filamentation modeの線形計算

問題設定

磁気シアを与えない： $B_y^{\text{eq}}/B_0 = 0$

$$\beta_e = 0.01, \quad \tau \equiv T_{0i}/T_{0e} = 0.5 \quad \sigma \equiv m_e/m_i = 1/1836, \\ \eta_e \equiv L_{n_{0e}}/L_{T_{0e}} = 1, \quad \nu_{ei}/\omega_{*e}(k_y = 0.1) = 1.5$$

GS2とAstroGKでそれぞれ計算

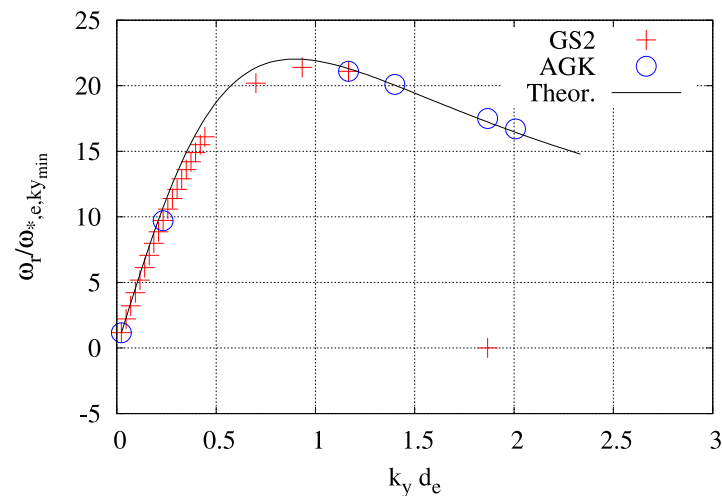
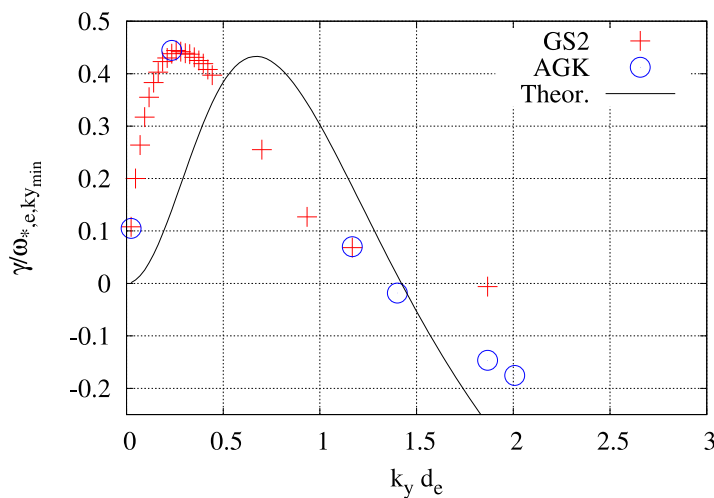
- GS2：Singleモードの固有値収束値を得る.
- AstroGK：boxグリッドにして，MTMと同様に計算（ただし，シアフリー）．磁場の時間発展から固有値を算出.

Self-filamentation modeの計算結果

分散関係
$$\left(1 - i\alpha_t \alpha_{tt} \frac{\omega_* T_{0e}}{\nu_{ei}}\right) \frac{\omega}{\nu_{ei}} = \frac{\omega_* n_{0e} + (1 + \alpha_t) \omega_* T_{0e}}{\nu_{ei}} - i\alpha_\eta (k_y d_e)^2 \left(1 - \frac{\alpha_i}{\alpha_\eta} \frac{i\omega}{\nu_{ei}}\right)$$

α 記号は衝突項由来の係数.

$$- i\alpha_\eta (k_y d_e)^2 \left(1 - \frac{\alpha_i}{\alpha_\eta} \frac{i\omega}{\nu_{ei}}\right)$$



非線形の問題設定

$$k_x^{\min} = k_y^{\min} = 0.2, \quad L_x = L_y = 10\pi, \quad n_x = n_y = 128,$$

理論計算の道のり

- Drakeらの手順を踏まえて、Drakeらの結果を再現できるのか？
→ レビュー的意味合い.
- 流体的なアプローチからDrakeらの結果を再現できるか？
→ 非線形項 $|\mathbf{k} \cdot \mathbf{B}_\perp|^2$ は、磁場とどの物理が相互作用しているか調べる.
- 磁場の圧縮効果を取り入れた輸送スケーリング則の導出. 流体的アプローチで.
 $\delta B_\parallel \times h$ の物理的意味を考える.

理論計算報告内容

Drakeらの結果を再現.

- 前回：ざっくりとした計算&途中で計算ミスがあった.
→丁寧に計算をし直した. しかし, Drakeらとは微妙な違いが生じる.

流体的なアプローチからのDrakeらの結果を再現.

- 前回：輸送方程式が必要&非線形項における ω の取り扱いについて.
- →非線形項 $|k \cdot B_{\perp}|^2$ は, 磁場と圧力揺動の相互作用から生じる.
摂動展開によって, 非線形項を考慮した固有値を導出→Drakeらの結果の再現ができない.

Nonlinear analysis of microtearing turbulence

Mitsuyoshi Yagyu

1 Gyrokinetic equation

A gyrokinetic equation is given by

$$\frac{\partial h_s}{\partial t} + v_{\parallel} \frac{B_y^{\text{eq}}(x)}{B_0} \frac{\partial h_s}{\partial Y_s} + \frac{1}{B_0} \{ \langle \chi \rangle_{\mathbf{R}s}, h_s \} = \frac{q_s f_{0s}}{T_{0s}} \left(\frac{\partial}{\partial t} + v_{*,s}^T(v) \frac{\partial}{\partial Y_s} \right) \langle \chi \rangle_{\mathbf{R}s} + C(h_s), \quad (1)$$

where

h_s can be decomposed into the first two lower-order moments and the other as [zocco+, 2011]

$$h_s = \left(\frac{q_s \phi}{T_{0s}} + \frac{\delta n_s}{n_{0s}} + \frac{2v_{\parallel} u_{\parallel,s}}{v_{\text{th},s}^2} \right) f_{0s} + h'_s, \quad (2)$$

and

$$\frac{\delta T_{\parallel,s}}{T_{0s}} = \frac{1}{n_{0s}} \int 2 \frac{v_{\parallel}^2}{v_{\text{th},s}^2} h'_s \mathrm{d}v \quad (3)$$

2 Fluid approach

We consider set of eigenvalue equations including the nonlinear term.

3 Review of Drake+, 1980

3.1 Basic equation

Consider the drift-kinetic equation of magnetic shear free for electrons. Furthermore, the electrostatic potential ϕ and the magnetic compressibility $\delta B_{\parallel}$ are ignored here. Then, (1) is represented by

$$\frac{\partial h_e}{\partial t} - \frac{v_{\parallel}}{B_0} \{ A_{\parallel}, h_e \} = -v_{\parallel} \frac{q_e f_{0e}}{T_{0e}} \left(\frac{\partial}{\partial t} + v_{*,e}^T(v) \frac{\partial}{\partial y} \right) A_{\parallel} + C(h_s). \quad (4)$$

The Lorentz collision operator is employed here, which is defined as

$$C(h_e) = \frac{\nu_{\text{ei}}(v)}{2} \frac{\partial}{\partial \xi} (1 - \xi^2) \frac{\partial}{\partial \xi} h_e, \quad (5)$$

where $\xi \equiv v_{\parallel}/v$ is the pitch angle variable.

Using the relation

$$\frac{v_{\parallel}}{B_0} \{ A_{\parallel}, h_e \} = -v_{\parallel} \frac{\mathbf{B}_{\perp}}{B_0} \cdot \nabla_{\perp} h_e, \quad (6)$$

the equation corresponding to Eq.(6) in [Drake+, 1980] is obtained as

$$\left(\frac{\partial}{\partial t} + v_{\parallel} \frac{\mathbf{B}_{\perp}}{B_0} \cdot \nabla_{\perp} - \frac{\nu_{\text{ei}}(v)}{2} \frac{\partial}{\partial \xi} (1 - \xi^2) \frac{\partial}{\partial \xi} \right) h_e = -v_{\parallel} \frac{q_e f_{0e}}{T_{0e}} \left(\frac{\partial}{\partial t} + v_{*,e}^T(v) \frac{\partial}{\partial y} \right) A_{\parallel}. \quad (7)$$

3.2 Perturbative expansion

Consider the perturbative expansion for h_e up to the third order in $\mathbf{B}_\perp/B_0 \sim h_e/f_{0e} \sim k_\perp A_\parallel/B_0 \sim \epsilon$, namely, $h_e = \epsilon h_{e,1} + \epsilon^2 h_{e,2} + \epsilon^3 h_{e,3} + \mathcal{O}(\epsilon^4)$. Separating (7) by order,

$$\epsilon^1 : \left(\frac{\partial}{\partial t} - \frac{\nu_{ei}(v)}{2} \frac{\partial}{\partial \xi} (1 - \xi^2) \frac{\partial}{\partial \xi} \right) h_{e,1} = -v_\parallel \frac{q_e f_{0e}}{T_{0e}} \left(\frac{\partial}{\partial t} + v_{*,e}^T(v) \frac{\partial}{\partial y} \right) A_\parallel, \quad (8)$$

$$\epsilon^2 : \left(\frac{\partial}{\partial t} - \frac{\nu_{ei}(v)}{2} \frac{\partial}{\partial \xi} (1 - \xi^2) \frac{\partial}{\partial \xi} \right) h_{e,2} = -v_\parallel \frac{\mathbf{B}_\perp}{B_0} \cdot \nabla_\perp h_{e,1}, \quad (9)$$

$$\epsilon^3 : \left(\frac{\partial}{\partial t} - \frac{\nu_{ei}(v)}{2} \frac{\partial}{\partial \xi} (1 - \xi^2) \frac{\partial}{\partial \xi} \right) h_{e,3} = -v_\parallel \frac{\mathbf{B}_\perp}{B_0} \cdot \nabla_\perp h_{e,2} \quad (10)$$

In the following, the subscripts 'e' and 'ei' are omitted.

3.2.1 On ϵ^1

Decompose h_1 and $A_\parallel$ into Fourier mode,

$$h_1 = \sum_{\mathbf{k}} h_{1,\mathbf{k}} e^{i(\mathbf{k} \cdot \mathbf{r} - \omega_{\mathbf{k}} t)}, \quad (11)$$

$$A_\parallel = \sum_{\mathbf{k}} A_{\parallel,\mathbf{k}} e^{i(\mathbf{k} \cdot \mathbf{r} - \omega_{\mathbf{k}} t)}, \quad (12)$$

to obtain

$$\left(-i\omega_{\mathbf{k}} - \frac{\nu}{2} \frac{\partial}{\partial \xi} (1 - \xi^2) \frac{\partial}{\partial \xi} \right) h_{1,\mathbf{k}} = i v_\parallel \frac{q f_0}{T_0} \left(\omega_{\mathbf{k}} - \omega_{*,k_y}^T(v) \right) A_{\parallel,\mathbf{k}}, \quad (13)$$

where $\omega_{*,k_y}^T(v) \equiv k_y v_*^T(v)$.

Expanding $h_{1,\mathbf{k}}$ by Legendre polynomials $h_{1,\mathbf{k}} = \sum_n h_{1,n}(v) P_n(\xi)$,

$$\sum_n \left(-i\omega_{\mathbf{k}} + \frac{\nu}{2} n(n+1) \right) h_{1,n}(v) P_n(\xi) = i v_\parallel \frac{q f_0}{T_0} \left(\omega_{\mathbf{k}} - \omega_{*,k_y}^T(v) \right) A_{\parallel,\mathbf{k}}, \quad (14)$$

and multiplying (14) by $P_m(\xi)$, the orthogonality of the Legendre polynomials yields

$$\text{LHS of (14)} = \sum_n \left(-i\omega_{\mathbf{k}} + \frac{\nu}{2} n(n+1) \right) h_{1,n}(v) \int_{-1}^1 P_n(\xi) P_m(\xi) d\xi = [\dots] h_{1,m}(v) \frac{2}{2m+1}, \quad (15)$$

$$\text{RHS of (14)} = i v_\parallel \frac{q f_0}{T_0} \left(\omega_{\mathbf{k}} - \omega_{*,k_y}^T(v) \right) A_{\parallel,\mathbf{k}} \int_{-1}^1 \xi P_m(\xi) d\xi, \rightarrow m = 1, \quad (16)$$

then,

$$h_{1,1} = i v_\parallel \frac{q f_0}{T_0} \frac{\omega_{\mathbf{k}} - \omega_{*,k_y}^T(v)}{\nu - i\omega_{\mathbf{k}}} A_{\parallel,\mathbf{k}}. \quad (17)$$

Therefore,

$$h_{1,\mathbf{k}} = h_{1,1} P_1(\xi) = i v_\parallel \frac{q f_0}{T_0} \frac{\omega_{\mathbf{k}} - \omega_{*,k_y}^T(v)}{\nu - i\omega_{\mathbf{k}}} A_{\parallel,\mathbf{k}}. \quad (18)$$

Substituting $q = -e$, (18) agrees with that of [Drake+, 1980].

3.2.2 On ϵ^2

Fourier mode decomposition:

$$h_2 = \sum_{\mathbf{k}_2} h_{2,\mathbf{k}_2} e^{i(\mathbf{k}_2 \cdot \mathbf{r} - \omega_{\mathbf{k}_2} t)}, \quad (19)$$

$$\mathbf{B}_\perp = \sum_{\mathbf{k}'} \mathbf{B}_{\perp,\mathbf{k}'} e^{i(\mathbf{k}' \cdot \mathbf{r} - \omega_{\mathbf{k}'} t)}, \quad (20)$$

$$h_1 = \sum_{\mathbf{k}} h_{1,\mathbf{k}} e^{i(\mathbf{k} \cdot \mathbf{r} - \omega_{\mathbf{k}} t)}. \quad (21)$$

From (9),

$$\sum_{\mathbf{k}_2} \left(-i\omega_{\mathbf{k}_2} - \frac{\nu}{2} \frac{\partial}{\partial \xi} (1 - \xi^2) \frac{\partial}{\partial \xi} \right) h_{2,\mathbf{k}_2} e^{i(\mathbf{k}_2 \cdot \mathbf{r} - \omega_{\mathbf{k}_2} t)} = -v_\parallel \sum_{\mathbf{k}_2 = \mathbf{k}' + \mathbf{k}} \left[\sum_{\mathbf{k}', \mathbf{k}} \frac{\mathbf{B}_{\perp,\mathbf{k}'} \cdot (i\mathbf{k})}{B_0} h_{1,\mathbf{k}} \right] e^{i(\mathbf{k}_2 \cdot \mathbf{r} - [\omega_{\mathbf{k}} + \omega_{\mathbf{k}'}] t)}, \quad (22)$$

thus,

$$\left(-i\omega_{\mathbf{k}_2} - \frac{\nu}{2} \frac{\partial}{\partial \xi} (1 - \xi^2) \frac{\partial}{\partial \xi} \right) h_{2,\mathbf{k}_2} = -iv_\parallel \sum_{\mathbf{k}', \mathbf{k}} \frac{\mathbf{k} \cdot \mathbf{B}_{\perp,\mathbf{k}'}}{B_0} h_{1,\mathbf{k}} e^{i\Delta\omega t}, \quad (23)$$

where $\Delta\omega \equiv \omega_{\mathbf{k}_2} - \omega_{\mathbf{k}} - \omega_{\mathbf{k}'}$. We assume $\Delta\omega = 0$ because for the long-time average,

$$\frac{1}{T} \int_0^T e^{i\Delta\omega t} dt = \frac{1}{T} \frac{e^{i\Delta\omega T} - 1}{i\Delta\omega} \xrightarrow{T \rightarrow \infty} 0. \quad (24)$$

Substituting (18) into (23),

$$\left(-i\omega_{\mathbf{k}_2} - \frac{\nu}{2} \frac{\partial}{\partial \xi} (1 - \xi^2) \frac{\partial}{\partial \xi} \right) h_{2,\mathbf{k}_2} = v_\parallel^2 \frac{qf_0}{T_0} \sum_{\mathbf{k}', \mathbf{k}} \frac{\mathbf{k} \cdot \mathbf{B}_{\perp,\mathbf{k}'}}{B_0} \frac{\omega_{\mathbf{k}} - \omega_{*,k_y}^T(v)}{\nu - i\omega_{\mathbf{k}}} A_{\parallel,\mathbf{k}} \quad (25)$$

$$= v^2 \xi^2 \frac{qf_0}{T_0} \sum_{\mathbf{k}', \mathbf{k}} \frac{\mathbf{k} \cdot \mathbf{B}_{\perp,\mathbf{k}'}}{B_0} \frac{\omega_{\mathbf{k}} - \omega_{*,k_y}^T(v)}{\nu - i\omega_{\mathbf{k}}} A_{\parallel,\mathbf{k}}. \quad (26)$$

Using Legendre polynomials and the orthogonality,

$$\left(-i\omega_{\mathbf{k}_2} + \frac{\nu}{2} m(m+1) \right) h_{2,m}(v) \frac{2}{2m+1} = v^2 \frac{qf_0}{T_0} \sum_{\mathbf{k}', \mathbf{k}} \frac{\mathbf{k} \cdot \mathbf{B}_{\perp,\mathbf{k}'}}{B_0} \frac{\omega_{\mathbf{k}} - \omega_{*,k_y}^T(v)}{\nu - i\omega_{\mathbf{k}}} A_{\parallel,\mathbf{k}} \int_{-1}^1 \xi^2 P_m(\xi) d\xi \quad (27)$$

$$= \dots \int_{-1}^1 \left(\frac{2}{3} P_2 P_m(\xi) + \frac{1}{3} P_0 P_m \right) d\xi \rightarrow m = 0, 2. \quad (28)$$

Since $h_{2,m}(v)$ decreases as m increases due to the effect of collisions, $m \geq 2$ is negligible here. Considering only $m = 0$ yields

$$h_{2,\mathbf{k}_2} = h_{2,0} P_0 = i \frac{v^2}{3\omega_{\mathbf{k}_2}} \frac{qf_0}{T_0} \sum_{\mathbf{k}', \mathbf{k}} \frac{\mathbf{k} \cdot \mathbf{B}_{\perp,\mathbf{k}'}}{B_0} \frac{\omega_{\mathbf{k}} - \omega_{*,k_y}^T(v)}{\nu - i\omega_{\mathbf{k}}} A_{\parallel,\mathbf{k}} \quad (29)$$

Due to the collisional condition $\omega/\nu \ll 1$,

$$h_{2,\mathbf{k}_2} = i \frac{v^2}{3} \frac{qf_0}{T_0} \sum_{\mathbf{k}', \mathbf{k}} \frac{\mathbf{k} \cdot \mathbf{B}_{\perp,\mathbf{k}'}}{B_0} A_{\parallel,\mathbf{k}} \frac{\omega_{\mathbf{k}} - \omega_{*,k_y}^T(v)}{\omega_{\mathbf{k}_2} \nu} \left(1 + i \frac{\omega_{\mathbf{k}_2}}{\nu} \right). \quad (30)$$

The sign disagree with Drake's result.

3.2.3 On ϵ^3

Fourier mode decomposition:

$$h_3 = \sum_{\mathbf{k}_3} h_{3,\mathbf{k}_3} e^{i(\mathbf{k}_3 \cdot \mathbf{r} - \omega_{\mathbf{k}_3} t)}, \quad (31)$$

$$\mathbf{B}_\perp = \sum_{\mathbf{k}''} \mathbf{B}_{\perp,\mathbf{k}''} e^{i(\mathbf{k}'' \cdot \mathbf{r} - \omega_{\mathbf{k}''} t)}, \quad (32)$$

$$h_2 = \sum_{\mathbf{k}_2} h_{2,\mathbf{k}_2} e^{i(\mathbf{k}_2 \cdot \mathbf{r} - \omega_{\mathbf{k}_2} t)} \quad (33)$$

From (10),

$$\sum_{\mathbf{k}_3} \left(-i\omega_{\mathbf{k}_3} - \frac{\nu}{2} \frac{\partial}{\partial \xi} (1 - \xi^2) \frac{\partial}{\partial \xi} \right) h_{3,\mathbf{k}_3} e^{i(\mathbf{k}_3 \cdot \mathbf{r} - \omega_{\mathbf{k}_3} t)} = -v_\parallel \sum_{\mathbf{k}_3 = \mathbf{k}'' + \mathbf{k}_2} \left[\sum_{\mathbf{k}'', \mathbf{k}_2} \frac{\mathbf{B}_{\perp,\mathbf{k}''}}{B_0} \cdot (i\mathbf{k}_2) h_{2,\mathbf{k}_2} \right] e^{i(\mathbf{k}_3 \cdot \mathbf{r} - [\omega_{\mathbf{k}_2} + \omega_{\mathbf{k}''}] t)}, \quad (34)$$

thus,

$$\left(-i\omega_{\mathbf{k}_3} - \frac{\nu}{2} \frac{\partial}{\partial \xi} (1 - \xi^2) \frac{\partial}{\partial \xi} \right) h_{3,\mathbf{k}_3} = -iv_\parallel \sum_{\mathbf{k}'', \mathbf{k}_2} \frac{\mathbf{k}_2 \cdot \mathbf{B}_{\perp,\mathbf{k}''}}{B_0} h_{2,\mathbf{k}_2} e^{i\Delta\omega t}, \quad (35)$$

where $\Delta\omega' \equiv \omega_{\mathbf{k}_3} - \omega_{\mathbf{k}_2} - \omega_{\mathbf{k}''} = 0$.

Substituting (30) into (35),

$$\left(-i\omega_{\mathbf{k}_3} - \frac{\nu}{2} \frac{\partial}{\partial \xi} (1 - \xi^2) \frac{\partial}{\partial \xi} \right) h_{3,\mathbf{k}_3} = v^2 v_\parallel \frac{qf_0}{3T_0} \sum_{\mathbf{k}'', \mathbf{k}_2} \frac{\mathbf{k}_2 \cdot \mathbf{B}_{\perp,\mathbf{k}''}}{B_0} \sum_{\mathbf{k}', \mathbf{k}} \frac{\mathbf{k} \cdot \mathbf{B}_{\perp,\mathbf{k}'}}{B_0} A_{\parallel,\mathbf{k}} \frac{\omega_{\mathbf{k}} - \omega_{*,k_y}^T(v)}{\omega_{\mathbf{k}_2} \nu} \left(1 + i \frac{\omega_{\mathbf{k}_2}}{\nu} \right). \quad (36)$$

Note that $\mathbf{k}'' + \mathbf{k}_2 = \mathbf{k}_3$ and $\mathbf{k}' + \mathbf{k} = \mathbf{k}_2$.

Using Legendre polynomials and the orthogonality,

$$\begin{aligned} \left(-i\omega_{\mathbf{k}_3} + \frac{\nu}{2} m(m+1) \right) h_{3,m}(v) \frac{2}{2m+1} &= v^3 \frac{qf_0}{3T_0} \sum_{\mathbf{k}'', \mathbf{k}_2} \sum_{\mathbf{k}', \mathbf{k}} \mathbf{k}_2 \cdot \mathbf{k} \frac{\mathbf{B}_{\perp,\mathbf{k}''} \cdot \mathbf{B}_{\perp,\mathbf{k}'}}{B_0^2} A_{\parallel,\mathbf{k}} \frac{\omega_{\mathbf{k}} - \omega_{*,k_y}^T(v)}{\omega_{\mathbf{k}_2} \nu} \left(1 + i \frac{\omega_{\mathbf{k}_2}}{\nu} \right) \\ &\times \int_{-1}^1 \xi P_m(\xi) d\xi \rightarrow m = 1. \end{aligned} \quad (37)$$

Therefore,

$$h_{3,\mathbf{k}_3} = h_{3,1} P_1(\xi) = \frac{v^2 v_\parallel}{\nu - i\omega_{\mathbf{k}_3}} \frac{qf_0}{3T_0} \sum_{\mathbf{k}'', \mathbf{k}_2} \sum_{\mathbf{k}', \mathbf{k}} \mathbf{k}_2 \cdot \mathbf{k} \frac{\mathbf{B}_{\perp,\mathbf{k}''} \cdot \mathbf{B}_{\perp,\mathbf{k}'}}{B_0^2} A_{\parallel,\mathbf{k}} \frac{\omega_{\mathbf{k}} - \omega_{*,k_y}^T(v)}{\omega_{\mathbf{k}_2} \nu} \left(1 + i \frac{\omega_{\mathbf{k}_2}}{\nu} \right) \quad (38)$$

$$= \frac{v^2 v_\parallel}{\nu^2} \frac{qf_0}{3T_0} \sum_{\mathbf{k}'', \mathbf{k}_2} \sum_{\mathbf{k}', \mathbf{k}} \mathbf{k}_2 \cdot \mathbf{k} \frac{\mathbf{B}_{\perp,\mathbf{k}''} \cdot \mathbf{B}_{\perp,\mathbf{k}'}}{B_0^2} A_{\parallel,\mathbf{k}} \frac{\omega_{\mathbf{k}} - \omega_{*,k_y}^T(v)}{\omega_{\mathbf{k}_2}} \left(1 + i \frac{\omega_{\mathbf{k}_2}}{\nu} \right) \left(1 + i \frac{\omega_{\mathbf{k}_3}}{\nu} \right) \quad (39)$$

3.2.4 Summary

The results of the perturbative expansion are listed below.

$$h_{1,\mathbf{k}} = i v_{\parallel} \frac{q f_0}{T_0} \frac{\omega_{\mathbf{k}} - \omega_{*,k_y}^T(v)}{\nu - i \omega_{\mathbf{k}}} A_{\parallel,\mathbf{k}}, \quad (40)$$

$$h_{2,\mathbf{k}} = i \frac{v^2}{3} \frac{q f_0}{T_0} \sum_{\mathbf{k}'+\mathbf{k}''=\mathbf{k}} \frac{\mathbf{k}' \cdot \mathbf{B}_{\perp,\mathbf{k}''}}{B_0} A_{\parallel,\mathbf{k}'} \frac{\omega_{\mathbf{k}'} - \omega_{*,k'_y}^T(v)}{\omega_{\mathbf{k}} \nu} \left(1 + i \frac{\omega_{\mathbf{k}}}{\nu}\right) \quad (41)$$

$$= i \frac{v^2}{3} \frac{q f_0}{T_0} \sum_{\mathbf{k}'+\mathbf{k}''=\mathbf{k}} \frac{\mathbf{k}' \cdot \mathbf{B}_{\perp,\mathbf{k}-\mathbf{k}'}}{B_0} A_{\parallel,\mathbf{k}'} \frac{\omega_{\mathbf{k}'} - \omega_{*,k'_y}^T(v)}{\omega_{\mathbf{k}} \nu} \left(1 + i \frac{\omega_{\mathbf{k}}}{\nu}\right), \quad (42)$$

$$h_{3,\mathbf{k}} = \frac{v^2 v_{\parallel}}{\nu^2} \frac{q f_0}{3 T_0} \sum_{\mathbf{k}'+\mathbf{k}''=\mathbf{k}} \sum_{\mathbf{p}'+\mathbf{p}''=\mathbf{k}'} \mathbf{k}'' \cdot \mathbf{p}'' \frac{\mathbf{B}_{\perp,\mathbf{k}'} \cdot \mathbf{B}_{\perp,\mathbf{p}'}}{B_0^2} A_{\parallel,\mathbf{p}''} \frac{\omega_{\mathbf{p}''} - \omega_{*,p''_y}^T(v)}{\omega_{\mathbf{k}''}} \left(1 + i \frac{\omega_{\mathbf{k}''}}{\nu}\right) \left(1 + i \frac{\omega_{\mathbf{k}}}{\nu}\right). \quad (43)$$

The expressions derived by Drake are

$$h_{1,\mathbf{k}} = -i v_{\parallel} \frac{e f_0}{T_0} \frac{\omega_{\mathbf{k}} - \omega_{*,k_y}^T(v)}{\nu - i \omega_{\mathbf{k}}} A_{\parallel,\mathbf{k}}, \quad (44)$$

$$h_{2,\mathbf{k}} = i \frac{v^2}{3} \frac{e f_0}{T_0} \sum_{\mathbf{k}'} \frac{\mathbf{k}' \cdot \mathbf{B}_{\perp,\mathbf{k}-\mathbf{k}'}}{B_0} A_{\parallel,\mathbf{k}'} \frac{\omega_{\mathbf{k}'} - \omega_{*,k'_y}^T(v)}{\omega_{\mathbf{k}} \nu}, \quad (45)$$

$$h_{3,\mathbf{k}} = \frac{v^2 v_{\parallel}}{\nu^2} \frac{e f_0}{3 T_0} \sum_{\mathbf{k}'} \frac{|\mathbf{k} \cdot \mathbf{B}_{\perp,\mathbf{k}'}|^2}{B_0^2} A_{\parallel,\mathbf{k}} \frac{\omega_{\mathbf{k}} - \omega_{*,k_y}^T(v) - [\omega_{\mathbf{k}'} - \omega_{*,k'_y}^T(v)]}{\omega_{\mathbf{k}} + \omega_{\mathbf{k}'}} \quad (46)$$

where $q = -e$.

4 Legendre polynomials expansion

Memo:

Legendre polynomials P_l satisfies

$$\frac{\partial}{\partial \xi} (1 - \xi^2) \frac{\partial}{\partial \xi} P_l(\xi) = -l(l+1) P_l(\xi). \quad (47)$$

The first three lower P_l are

$$P_0(\xi) = 1, \quad P_1(\xi) = \xi, \quad P_2(\xi) = \frac{1}{2}(3\xi^2 - 1). \quad (48)$$

Legendre expansion is given by

$$h(\xi) = \sum_l h_l P_l(\xi). \quad (49)$$

Orthogonality

$$\int_{-1}^1 P_l(\xi) P_m(\xi) d\xi = \frac{2}{2l+1} \delta_{l,m}, \quad (50)$$

where $\delta_{l,m}$ is the Dirac's delta function.

5 Lorentz collision operator

Lorentz collision operator is given by

$$C_L(h) = \frac{\nu(v)}{2} \frac{\partial}{\partial \xi} (1 - \xi^2) \frac{\partial}{\partial \xi} h, \quad (51)$$

where $\xi = v_{\parallel}/v$ is the pitch angle variable, $\nu(v)$ is the pitch angle scattering frequency dependent of the velocity and $h = h(v_{\parallel}, v_{\perp})$ is the velocity distribution function. Lorentz collision operator considers only collisions of light particles with heavy particles, neglecting energy loss, that is, the particle energy is not conserved.

5.1 $v_{\parallel}^2$ moment of (51)

$v_{\parallel}^2$ moment is

$$I \equiv \int v_{\parallel}^2 C_L(h) d\mathbf{v} \quad (52)$$

$$= \int v^2 \xi^2 \frac{\nu}{2} \frac{\partial}{\partial \xi} (1 - \xi^2) \frac{\partial}{\partial \xi} h d\mathbf{v}. \quad (53)$$

Transforming the velocity coordinate $(v_{\parallel}, v_{\perp})$ to (v, ξ) using Jacobian,

$$d\mathbf{v} = 2\pi dv_{\parallel} v_{\perp} = 2\pi v_{\perp} |J(v, \xi)| dv d\xi = 2\pi v^2 dv d\xi, \quad (54)$$

where $v_{\perp} = v\sqrt{1 - \xi^2}$ and $|J(v, \xi)| = v/\sqrt{1 - \xi^2}$.

Using Legendre polynomials expansion,

$$I = -2\pi \sum_l \frac{l(l+1)}{2} \int_0^{\infty} v^4 \nu(v) h_l(v) dv \int_{-1}^1 \xi^2 P_l(\xi) d\xi \quad (55)$$

$$= -2\pi \sum_l \frac{l(l+1)}{2} \int_0^{\infty} v^4 \nu(v) h_l(v) dv \int_{-1}^1 \left(\frac{2}{3} P_2(\xi) P_n(\xi) + \frac{1}{3} P_0(\xi) P_n(\xi) \right) d\xi. \quad (56)$$

$l = 0, 2$ survive due to the orthogonality of Legendre polynomials defined in (50), but if $l = 0$, $I = 0$. Therefore,

$$\int v_{\parallel}^2 C_L(h) d\mathbf{v} = -\frac{8}{5}\pi \int_0^{\infty} v^4 \nu(v) h_2(v) dv \quad (57)$$

If collisions between particles are sufficiently strong, h_l with higher l is negligible. In [Drake+, 1980], h_l for $l \geq 2$ is neglected in the collisional regime, thereby satisfying $\int v_{\parallel}^2 C_L(h) d\mathbf{v} = 0$.

①

2.7.17 Gk eqn (S: 磁気場) , $h_s = \sigma f_s + \frac{\sigma \tau}{\tau_0} f_{0s}$

$$\begin{aligned} \frac{\partial h_s}{\partial t} + u_{||} \frac{B_y^e}{B_0} \frac{\partial}{\partial r_s} h_s + \frac{1}{B_0} \{ \langle \chi \rangle_{R_s} h_s \} \\ = \frac{\sigma \tau f_{0s}}{\tau_0} \left[u_{*nos} \left(1 + \left[\frac{u_{||}^2}{2u_{th}^2} - \frac{3}{2} \right] \eta_s \right) \frac{\partial}{\partial r_s} \langle \chi \rangle_{R_s} + \frac{\partial}{\partial t} \langle \chi \rangle_{R_s} \right] \\ + C(h_s) \quad \text{--- (*)} \end{aligned}$$

$\frac{B_y^e}{B_0}(\chi)$: Magnetic shear term

$$\langle \chi \rangle_{R_s} = \sum_k \left(\tilde{J}_0(\alpha_s) (\tau_0 - u_{||} A_{||,k}) + \frac{\tau_0}{\sigma} \frac{\tilde{J}_1(\alpha_s)}{\alpha_s} \frac{u_{||}^2}{2u_{th}^2} \frac{\partial B_{||,k}}{\partial r_s} \right) e^{i\phi_0 R_s}$$

$$u_{*nos} \equiv -\frac{\tau_0}{\sigma B_0 L_{nos}} \quad , \quad \eta_s \equiv \frac{L_{\tau_0}^{-1}}{L_{nos}^{-1}}$$

$$f_{0s} = \frac{n_{0s}}{(\sqrt{\pi} 2u_{th})^3} e^{-\frac{u_{||}^2}{2u_{th}^2}}$$

Zocco et al, 2011 を参照し (以下a) 係を使う

$$h_s = \left(\frac{\sigma \tau}{\tau_0} + \frac{\sigma n_s}{n_{0s}} + \frac{2 u_{||} u_{||,s}}{2u_{th}^2} \right) f_{0s} + h'_s \quad \text{--- (1)}$$

h'_s : 2次以上 $u_{||}$ の項を含む。

$$\left\{ \begin{aligned} \frac{\sigma \tau_{||,e}}{\tau_{0e}} &= \frac{1}{n_{0e}} \int 2 \frac{u_{||}^2}{2u_{th}^2} h'_e d\chi \\ \int (u_{||}) h'_s &= 0. \end{aligned} \right.$$

e^- の平均速度 $\hat{z}$ 方向 (シフト + 非等価性による e^- の平均速度)

(2)

$B_{\perp} S_S \ll 1$, e^- FLR 効果は無視

$$\Rightarrow \langle A \rangle_{Re} = A = \sum_k \left(\varphi_k - u_{\parallel} A_{\parallel} + \frac{T_{0s}}{B_0} \frac{u_{\perp}^2}{2 n_{\perp}^2} \frac{\partial B_{\parallel}}{\partial z} \right) e^{i k \cdot r}$$

$$= \varphi_k - u_{\parallel} A_{\parallel} + \frac{T_{0s}}{B_0} \frac{u_{\perp}^2}{2 n_{\perp}^2} \frac{\partial B_{\parallel}}{\partial z}$$

(*) Gkeyn $a_{\parallel}$ の意味と z 軸 (1) $a_{\parallel}$ の係数

$$\int u_{\parallel} \frac{\partial h_e}{\partial t} dz = \frac{\partial}{\partial t} \int u_{\parallel} \left[\left(\frac{\partial \varphi}{\partial z} + \frac{\partial h_e}{\partial u_{\parallel}} + \frac{2 u_{\parallel} u_{\parallel P}}{2 n_{\perp}^2} \right) f_{0e} + h_e' \right] dz$$

$$= \frac{\partial}{\partial t} u_{\parallel} e \cdot 2 \int \frac{u_{\parallel}^2}{2 n_{\perp}^2} f_{0e} dz$$

$$= n_{0e} \frac{\partial}{\partial t} u_{\parallel} e$$

$$\int u_{\parallel} u_{\parallel} \frac{B_y^2(z)}{B_0} \frac{\partial}{\partial z} h_e dz$$

$$= u_{\parallel}^2 \frac{B_y^2}{B_0} \frac{\partial}{\partial z} \left[\int \left(\frac{\partial \varphi}{\partial z} + \frac{\partial h_e}{\partial u_{\parallel}} \right) \frac{1}{2} f_{0e} dz + \int u_{\parallel}^2 h_e' dz \right]$$

$$= \frac{n_{0e} T_{0e}}{m_e} \frac{B_y^2}{B_0} \frac{\partial}{\partial z} \left(\frac{\partial \varphi}{\partial z} + \frac{\partial h_e}{\partial u_{\parallel}} + \frac{\partial T_{\parallel P}}{T_{0e}} \right)$$

先右辺を計算

$$\vec{z}^2 \equiv \frac{z^2}{z_{th}^2}$$

$$\int z_{ir} \frac{\delta f_{ir}}{T_{oe}} \left(1 - \frac{3}{2} \gamma_e + \hat{z}^2 \gamma_e \right) z_{thoe} \frac{\partial}{\partial y} \left(\varphi - z_{ir} A_{ir} + \frac{T_{oe}}{T_{ie}} \frac{z_{ir}^2}{z_{th}^2} \frac{\partial B_{ir}}{\partial z} \right) dz$$

$$= - \frac{\gamma_e}{T_{oe}} z_{thoe} \left(1 - \frac{3}{2} \gamma_e \right) \frac{\partial}{\partial y} A_{ir} z_{th}^2 \int \hat{z}_{ir}^2 f_{oe} dz = \frac{n_{oe}}{2}$$

$$- \frac{\gamma_e}{T_{oe}} z_{thoe} \gamma_e \frac{\partial}{\partial y} A_{ir} z_{th}^2 \int \hat{z}_{ir}^2 \hat{z}^2 f_{oe} dz = \frac{5}{4} n_{oe}$$

$$= - \frac{\gamma_e n_{oe}}{m_e} z_{thoe} \left[1 - \frac{3}{2} \gamma_e + \frac{5}{2} \gamma_e \right] \frac{\partial}{\partial y} A_{ir}$$

$$= - \frac{\gamma_e n_{oe}}{m_e} z_{thoe} (1 + \gamma_e) \frac{\partial}{\partial y} A_{ir}$$

$$\int z_{ir} \frac{\delta f_{ir}}{T_{oe}} \frac{\partial}{\partial t} \chi dz$$

$$= - \frac{\gamma_e}{T_{oe}} \frac{\partial}{\partial t} A_{ir} z_{th}^2 \int \hat{z}_{ir}^2 f_{oe} dz$$

$$= - \frac{\gamma_e n_{oe}}{m_e} \frac{\partial}{\partial t} A_{ir}$$

$\vec{B} \cdot \nabla T_e$ (A_{ir}^{oe})
 $\vec{B} \times \nabla \approx \rho \hat{e}_\theta$ 含意は $\vec{E} \times \nabla$

$$\int z_{ir} C(h_e) dz \approx - \frac{\gamma_e n_{oe}}{m_e} \gamma \hat{z}_{ir} - \alpha_{te} \frac{n_e}{n_e} \nabla_{\parallel} T_e + \alpha_{te} \frac{1}{v_{ei} \partial t} \left(\frac{n_e \nabla_{\parallel} T_e}{m_e} \right)$$

(Hassam 1980 の collision 項 由来の 1 項を 2 項に 1 項を 2 項に)

② 1 項を 2 項に

非線形項の $z_{ir} z_{\perp}$ について

$$\begin{aligned} & \int z_{ir} \frac{1}{B_0} \left[\kappa_r h_e \right] d\mathbf{r} \\ &= \frac{1}{B_0} \left[\int z_{ir} \left(\frac{\partial \varphi}{\partial x} \frac{\partial}{\partial y} h_e - \frac{\partial}{\partial y} \varphi \frac{\partial}{\partial x} h_e \right) d\mathbf{r} \right. \\ & \quad - \int z_{ir}^2 \left(\frac{\partial}{\partial x} A_{ir} \frac{\partial}{\partial y} h_e - \frac{\partial}{\partial y} A_{ir} \frac{\partial}{\partial x} h_e \right) d\mathbf{r} \\ & \quad \left. + \frac{T_{oe}}{g B_0} \int z_{ir} \tilde{z}_{\perp}^2 \left(\frac{\partial}{\partial x} \delta B_r \frac{\partial}{\partial y} h_e - \frac{\partial}{\partial y} \delta B_r \frac{\partial}{\partial x} h_e \right) d\mathbf{r} \right] \end{aligned}$$

$$(1) : h_e = \left(\frac{\delta \varphi}{T_{oe}} + \frac{\delta h_e}{n_{oe}} + \frac{2 z_{ir} u_{ir,e}}{2 + h_e^2} \right) f_{oe} + h_e' \in \mathbb{R} \cup \mathbb{Z}$$

$$\begin{aligned} (7.3') &= \frac{1}{B_0} \left[\int \frac{\partial \varphi}{\partial x} \frac{\partial}{\partial y} u_{ir,e} \cdot 2 z_{ir}^2 f_{oe} d\mathbf{r} - \int \dots d\mathbf{r} \right. && \text{ } \varphi \times h_e \text{ term} \\ & \quad - \int \frac{\partial}{\partial x} A_{ir} \frac{\partial}{\partial y} \left[\left(\frac{\delta \varphi}{T_{oe}} + \frac{\delta h_e}{n_{oe}} \right) z_{ir}^2 f_{oe} + z_{ir}^2 h_e' \right] d\mathbf{r} - \int \dots d\mathbf{r} && \text{ } A_{ir} \times h_e \text{ term} \\ & \quad + \frac{T_{oe}}{g B_0} \int \frac{\partial}{\partial x} \delta B_r \frac{\partial}{\partial y} \left[u_{ir,e} \left(2 \tilde{z}_{ir}^2 \tilde{z}_{\perp}^2 f_{oe} \right) + z_{ir} \tilde{z}_{\perp}^2 h_e' \right] d\mathbf{r} && \text{ } \delta B_{ir} \times h_e \text{ term} \\ & \quad \left. - \int \dots d\mathbf{r} \right] \end{aligned}$$

(5)

φ x he term (1st term + 2nd term)

$$1^{st} \text{ term} = \frac{\partial}{\partial x} \varphi \frac{\partial}{\partial y} u_{||} \left[2 \int z_{||}^2 f_0 dz \right]_{n_0}$$

$$= n_0 \frac{\partial}{\partial x} \varphi \frac{\partial}{\partial y} u_{||}$$

$$\varphi \text{ x he term} = \frac{n_0}{B_0} \oint \varphi, u_{||} = - \frac{n_0 u_E \cdot \nabla_{\perp} u_{||}}{c}$$

$$u_E \equiv \frac{1}{B_0} \left(- \frac{\partial \varphi}{\partial x} \right)$$

$A_{||}$ x he term

$$1^{st} \text{ term} = \frac{\partial}{\partial x} A_{||} \frac{\partial}{\partial y} \left[\left(\frac{\delta \varphi}{T_0} + \frac{\delta n}{n_0} \right) \int z_{||}^2 f_0 dz + \int z_{||}^2 h_0' dz \right]$$

$$= \frac{n_0}{2} z_{||}^2 \frac{\partial}{\partial x} A_{||} \frac{\partial}{\partial y} \left[\left(\frac{\delta \varphi}{T_0} + \frac{\delta n}{n_0} \right) \frac{2}{n_0} \int z_{||}^2 f_0 dz + \frac{1}{n_0} \int 2 z_{||}^2 h_0' dz \right]$$

$$= \frac{n_0 T_0}{m} \frac{\partial}{\partial x} A_{||} \frac{\partial}{\partial y} \left[\frac{\delta \varphi}{T_0} + \frac{\delta n}{n_0} + \frac{\delta T_{||}}{T_0} \right]$$

$A_{||}$ x he term

$$= - \frac{1}{B_0} \frac{n_0 T_0}{m} \left[\oint A_{||} \varphi + \oint A_{||} \left(\frac{\delta n}{n_0} + \frac{\delta T_{||}}{T_0} \right) \right]$$

$$= \frac{n_0 T_0}{m} \left(\frac{1}{B_0} \oint \varphi A_{||} - \frac{1}{B_0} \oint A_{||} \left(\frac{\delta n}{n_0} + \frac{\delta T_{||}}{T_0} \right) \right)$$

$$= \frac{n_0 T_0}{m c} \left(- u_E \cdot \nabla_{\perp} A_{||} + \frac{\hat{B}}{B_0} \cdot \nabla_{\perp} \left[\frac{\delta n}{n_0} + \frac{\delta T_{||}}{T_0} \right] \right)$$

$$\frac{\hat{B}}{B_0} \cdot \nabla_{\perp} = - \frac{1}{B_0} \oint A_{||} f$$

$$A_{||}^{\text{avg}} \text{ is } \oint A_{||} f \text{ } \longrightarrow \hat{B} \equiv \frac{\hat{B}}{B_0} \text{ } \text{r.t.}$$

⑥

 $\delta B_{\perp} \times h_e$ term

1st term

$$= \frac{T_0}{g B_0} \frac{\partial \delta B_{\perp}}{\partial x} \frac{\partial}{\partial y} \left[u_{\parallel} \underbrace{\int 2 \tilde{z}_{\perp}^2 \tilde{z}_{\perp}^2 f_0 d\mathbf{r}}_{n_0} + \int \tilde{z}_{\perp} \tilde{z}_{\perp}^2 h'_e d\mathbf{r} \right]$$

$$= \frac{n_0 T_0}{g B_0} \left[\frac{\partial \delta B_{\perp}}{\partial x} \frac{\partial}{\partial y} u_{\parallel} + \frac{\partial \delta B_{\perp}}{\partial x} \frac{\partial}{\partial y} \left(\frac{2\pi\hbar}{n_0} \int \tilde{z}_{\perp} \tilde{z}_{\perp}^2 h'_e d\mathbf{r} \right) \right]$$

 $\delta B_{\perp} \times h_e$ term

$$M_3 \equiv \frac{1}{n_0} \int \tilde{z}_{\perp} \tilde{z}_{\perp}^2 h'_e d\mathbf{r}$$

$$= \frac{n_0 T_0}{g B_0^2} \left[\int \delta B_{\perp} u_{\parallel} + \int \delta B_{\perp} 2\pi\hbar M_3 \right]$$

$$= \frac{n_0 T_0}{g B_0^2} (\nabla_{\perp} \times \tilde{\mathbf{B}}) \cdot \nabla_{\perp} (u_{\parallel} + 2\pi\hbar M_3)$$

$$= \frac{n_0 T_0}{g B_0^2} \cdot \mu_0 \hat{\mathbf{j}}_{\perp} \cdot \nabla_{\perp} (u_{\parallel} + 2\pi\hbar M_3)$$

$$\hat{\mathbf{j}}_{\perp} \cdot \nabla_{\perp} = \frac{1}{\mu_0} \int \delta B_{\perp}$$

$$= - \frac{1}{2g} \beta \hat{\mathbf{j}}_{\perp} \cdot \nabla_{\perp} (u_{\parallel} + 2\pi\hbar M_3)$$

$$\nabla_{\perp} \times \mathbf{B} = \mu_0 \hat{\mathbf{j}}$$

$$(\nabla_{\perp} \times \mathbf{B}) \cdot \nabla_{\perp} = \mu_0 \hat{\mathbf{j}}_{\perp} \cdot \nabla_{\perp}$$

式 7.2

$Z_{11} E - f_{11} \propto \tau^-$ Glc 等々

$$n_{oe} \frac{\partial}{\partial t} U_{11e} + \frac{n_{oe} T_{oe}}{m_e} \frac{B_y^{pe}}{B_0} \left[\frac{\partial}{\partial y} \left(\frac{g_e \varphi}{T_{oe}} \right) + \frac{\partial}{\partial y} \left[\frac{\delta n_{oe}}{n_{oe}} + \frac{\delta T_{11e}}{T_{oe}} \right] \right]$$

$$- n_{oe} U_E \cdot \nabla_{\perp} \left(U_{11e} + \frac{g_e}{v_{te}} A_{11} \right)$$

$$+ \frac{n_{oe} T_{oe}}{m_e} \hat{H} \cdot \nabla_{\perp} \left[\frac{\delta n_{oe}}{n_{oe}} + \frac{\delta T_{11e}}{T_{oe}} \right]$$

$$+ \frac{1}{2 g_e} \beta_e \hat{y}_{\perp} \cdot \nabla_{\perp} \left(U_{11e} + Z_{11} M_{e3} \right)$$

$$= - \frac{g_e n_{oe}}{m_e} \left(Z_{11} n_{oe} (1 + \eta_e) \frac{\partial}{\partial y} A_{11} + \frac{\partial}{\partial t} A_{11} \right)$$

$$- \frac{g_e n_{oe}}{m_e} \eta \hat{y}_{\parallel} - \alpha_t \frac{n_{oe}}{m_e} \hat{H} \cdot \nabla_{\perp} T_e + \frac{\alpha_{te}}{m_e k_B} \frac{\partial}{\partial t} (n_{oe} \hat{H} \cdot \nabla_{\perp} T_e) - (*)'$$

物理的にこの
形式で書く。

(*)' x me 式 (1) 整理する

$\hat{H} \cdot \hat{H} = \hat{H}^2$

$$m_e n_{oe} \left(\frac{\partial}{\partial t} U_{11e} - U_E \cdot \nabla_{\perp} U_{11e} \right)$$

$$= - g_e n_{oe} \left(\frac{\partial}{\partial t} A_{11} + \frac{B_y^{pe}}{B_0} \frac{\partial}{\partial y} \varphi - U_E \cdot \nabla_{\perp} A_{11} \right)$$

$$- g_e n_{oe} Z_{11} n_{oe} (1 + \eta_e) \frac{\partial}{\partial y} A_{11}$$

$$- n_{oe} T_{oe} \left[\frac{B_y^{pe}}{B_0} \frac{\partial}{\partial y} \left(\frac{\delta n_{oe}}{n_{oe}} + \frac{\delta T_{11e}}{T_{oe}} \right) + \hat{H} \cdot \nabla_{\perp} \left(\frac{\delta n_{oe}}{n_{oe}} + \frac{\delta T_{11e}}{T_{oe}} \right) \right]$$

$$- g_e n_{oe} \eta \hat{y}_{\parallel} - \alpha_t n_{oe} \hat{H} \cdot \nabla_{\perp} T_e + \frac{\alpha_{te}}{m_e k_B} \frac{\partial}{\partial t} (n_{oe} \hat{H} \cdot \nabla_{\perp} T_e)$$

$$+ 2 \frac{m_e}{g_e} \beta_e \hat{y}_{\perp} \cdot \nabla_{\perp} \left(U_{11e} + Z_{11} M_{e3} \right) - (*)$$

(8)

Drake ϵ_r (180) と同じ設定で計算

$$\left\{ \begin{array}{l} \nabla \cdot \mathbf{E} = 0 \\ \nabla \cdot \mathbf{B} = 0 \end{array} \right.$$

 $\left\{ \begin{array}{l} \nabla \cdot \mathbf{B} = 0 \\ \nabla \cdot \mathbf{E} = 0 \end{array} \right.$
一般に $\mathbf{E} = -\nabla \phi$ である、 e^- の α subscript は省略

$$n_0 m \frac{\partial v_{||}}{\partial t} = -\gamma n_0 \left(\frac{\partial}{\partial t} A_{||} + Z_{*n} (1 + \gamma_e) \frac{\partial}{\partial y} A_{||} \right)$$

$$- \gamma n_0 \gamma \hat{J}_{||} - \alpha_k n_0 \hat{\mathbf{B}} \cdot \nabla_{\perp} T_{oe} + \alpha_{th} \frac{n_0}{2} \frac{\partial}{\partial t} (\hat{\mathbf{B}} \cdot \nabla_{\perp} T_{oe})$$

$$- n_0 T_0 \frac{B_{\perp}}{B_0} \cdot \nabla_{\perp} \left[\frac{\delta n}{n_0} + \frac{\delta T_{||}}{T_0} \right]$$

 $\int \mathbf{Z}_{*n} C(h) dh$
 対応する項

$$\gamma = \alpha_3 d^2 L \mu_0$$

$$\mu_0 \hat{J}_{||} = -\nabla_{\perp}^2 A_{||}$$

$$\hat{\mathbf{B}} \cdot \nabla_{\perp} T_{oe} = \gamma Z_{*T} \frac{\partial}{\partial y} A_{||}$$

$$Z_{*T} \equiv -\frac{T_0}{\gamma B_0} L_T^{-1}$$

$$\frac{\partial}{\partial t} v_{||} = -\frac{\gamma}{m} \left[\frac{\partial}{\partial t} + \left[Z_{*n} + (1 + \alpha_k) Z_{*T} \right] \frac{\partial}{\partial y} - \frac{\alpha_{th} Z_{*T}}{L} \frac{\partial}{\partial t} \frac{\partial}{\partial y} - \frac{\gamma}{\mu_0} \nabla_{\perp}^2 \right] A_{||}$$

$$- \frac{1}{2} Z_{*h}^2 \frac{B_{\perp}}{B_0} \cdot \nabla_{\perp} \left[\frac{\delta n}{n_0} + \frac{\delta T_{||}}{T_0} \right]$$

L (8)

④

⑤ 7-11エートに合格.

$$\psi = \sum_k \psi_k e^{i(k \cdot r - \omega_k t)}$$

$$\sum_k [-i\omega_k] u_{1r,k} e^{i(kr - \omega_k t)}$$

$$W_{\text{th},T} = \Delta Z_{\text{th},T}$$

$$w^* \equiv w^*h + (1+\alpha t)w^*T$$

$$= -\frac{\hbar}{m} \sum_k \left[-i\omega_R + i\omega_k - \alpha_{ff} + \frac{\omega_k^T}{v} \omega_R + \alpha_\eta \mathcal{L}(kd)^2 \right] \times A_{kR} e^{i(k_0 R - \omega_k t)}$$

$$-\frac{1}{2} \hbar^2 \sum_{\mathbf{k}=\mathbf{k}'+\mathbf{k}''} \sum_{\mathbf{k}'} \sum_{\mathbf{k}''} \frac{B_{\mathbf{k},\mathbf{k}'}}{B_0} \cdot (-i\hbar'') \left[\frac{\delta H_{\mathbf{k},\mathbf{k}''}}{\hbar_0} + \frac{\delta T_{\mathbf{k},\mathbf{k}''}}{T_0} \right] \times e^{i(\mathbf{k} \cdot \mathbf{r} - [\omega_{\mathbf{k}'} + \omega_{\mathbf{k}''}]t)}$$

$$i\omega_R U_{r,k} = -\frac{\mathcal{E}}{m} \left[i\omega_R - i\omega_* + \alpha_{th} \omega_* T \left(\frac{\omega_R}{L} \right) - \alpha_y L (kd)^2 \right] A_{r,k} \\ + \frac{1}{2} Z_{th}^2 \sum_{\substack{k' \neq k'' \\ |k' + k''| = |k|}} (iH'') \cdot \frac{B_{r,k'}}{B_0} \left[\frac{\delta n_{r,b}''}{n_0} + \frac{\delta T_{r,b}''}{T_0} \right] e^{i\Delta\omega t}$$

(2)

定常乱流に於ける長管内平均流速と

$$= \frac{1}{T} \frac{e^{i\Delta\omega T} - 1}{i\Delta\omega}$$

$$T \rightarrow \infty \Rightarrow 0 \rightarrow \Delta W \neq 0$$

5.2 $\Delta W = 0$ ㄷ ㄱ.

(10)

運動方程式と連続方程式 ($B_z^{\text{rf}} = 0, \varphi = 0, B_{\parallel} = 0$)

$$\left\{ \begin{aligned} \frac{\partial}{\partial t} \left(\frac{\delta T_r}{T_0} + \frac{\delta n}{n_0} \right) + 3 \tilde{B}_0 \cdot \nabla_{\perp} u_{\parallel} + \tilde{B}_0 \cdot \nabla_{\perp} 2\pi h \hat{M}_{3r} &= \frac{1}{n_0} \int 2 \tilde{A}_{\parallel}^2 C(h) d^2 h \\ \left(\hat{M}_{3r} \equiv \frac{1}{n_0} \int 2 \frac{h_{\parallel}^3}{2\pi h^3} h' d^2 h \right) & \quad \text{今は } \frac{u}{c} \gg 1 \\ & \quad \text{よって } 0 \\ & \quad \text{P1 参考 57} \\ \frac{\partial}{\partial t} \frac{\delta n}{n_0} + \tilde{B}_0 \cdot \nabla_{\perp} u_{\parallel} &= 0 \end{aligned} \right.$$

運動方程式 57

$$-i\omega_k \left[\frac{\delta n_{rk}}{n_0} + \frac{\delta T_{rk}}{T_0} \right] = - \sum_{\substack{k', k'' \\ k' + k'' = k}} (i k'') \cdot \frac{B_{\perp k'}}{B_0} \left(3 u_{\parallel k''} + 2\pi h \hat{M}_{rk''} \right)$$

簡単のため $\hat{M}_{rk''}$ を $\propto \delta(k'')$ と近似

$$\begin{aligned} \frac{\delta n_{rk}}{n_0} + \frac{\delta T_{rk}}{T_0} &= \sum_{\substack{k', k'' \\ k' + k'' = k}} \frac{3 k''}{\omega_k} \cdot \frac{B_{\perp k'}}{B_0} u_{\parallel k''} \\ &= \frac{8}{m} \sum \frac{3 k''}{\omega_k} \cdot \frac{B_{\perp k'}}{B_0} (k'' d)^2 A_{rk''} \end{aligned}$$

$$(u_{\parallel} = -\frac{8}{m} d e^2 \nabla_{\perp}^2 A_{\parallel})$$

1
(3)

(2) & (3) 代入する

$$i\omega_k \left[\frac{\delta}{m} (kd)^2 \right] A_{rk} = -\frac{\delta}{m} \left[i\omega_k - i\omega_* + \alpha_m \omega_* \tau \left(\frac{\omega_k}{L} \right) - \alpha_g L (kd)^2 \right] A_{rk} \\ + \frac{1}{2} \frac{\delta}{m} z_{th}^2 \sum_{k' \neq k} \frac{(i\hbar'') \cdot B_{rk'}}{B_0} \left[\sum_{p' \neq p} \frac{3p'}{\omega_{p'}} \cdot \frac{B_{rp'}}{B_0} (p'd)^2 A_{rp'} \right]$$

$\therefore \frac{\delta}{m} A_{rk} L^2$, 左辺の z_{th}^2 と ω_k は L^2 オーダー

$$\frac{\omega_k}{L} \left(1 + (kd)^2 - i\alpha_m \frac{\omega_* \tau}{L} \right) = \frac{\omega_*}{L} - i\alpha_g (kd)^2 \\ + \frac{3}{2} \left(\frac{z_{th}}{L} \right)^2 \sum_{k' \neq k} \sum_{p' \neq p} \frac{\hbar'' \cdot p'}{\omega_{p'}} \frac{B_{rk'} \cdot B_{rp'}}{B_0^2} (p'd)^2 \frac{A_{rp'}}{A_{rk}} \\ \frac{\omega_{p'}}{L} \left[\frac{\hbar'' \cdot p' z_{th}^2}{(\omega_{p'})^2} \cdot (p'd)^2 \frac{B_{rk'} \cdot B_{rp'}}{B_0^2} \right] \\ \propto \tau^{-2} \propto L^2$$

$$\text{今, } \frac{\omega}{k L z_{th}} = \frac{\omega}{\omega_*} \frac{\omega_*}{k L z_{th}}$$

$$\sim \frac{Se}{L_n} \sim \epsilon$$

$$\therefore \frac{\hbar'' \cdot p' z_{th}^2}{(\omega_{p'})^2} \sim \epsilon^{-2} \Rightarrow (p'd)^2 \sim \epsilon^2$$

(2)

$$\frac{\delta - \delta' - \delta''}{\omega} \sim \frac{\omega^*}{\omega} \sim kd \sim \frac{\omega}{k z_{th}} \sim \epsilon \ll 1$$

$$\text{したがって} \quad \frac{\omega}{\omega} = \frac{\omega_0}{\omega} + \epsilon \frac{\omega_1}{\omega} + \epsilon^2 \frac{\omega_2}{\omega} \quad \text{と展開する.}$$

$$\left(\frac{\omega_0}{\omega} + \frac{\omega_1}{\omega} + \frac{\omega_2}{\omega} \right) \left[1 + (kd)^2 - i \alpha_m \frac{\omega^*}{\omega} \right] = \frac{\omega^*}{\omega} - i \alpha_y (kd)^2$$

$$+ \frac{3}{2} \sum_{k'+k''=k} \sum_{p'+p''=p} \left[\frac{k'' \cdot p'' z_{th}^2}{(\omega_0'' + \omega_1'' + \omega_2'')^2} \left[\frac{\omega_0''}{\omega} + \frac{\omega_1''}{\omega} + \frac{\omega_2''}{\omega} \right] \right]$$

$$\times \frac{B_{\perp k'} \cdot B_{\perp p'}}{B_0^2} (p'' d)^2 \frac{A_{rp''}}{A_{rk}}$$

$$(\text{Nonlinear term}) = \frac{3}{2} \sum \sum \frac{k'' \cdot p'' z_{th}^2}{(\omega_0'')^2} \left[1 - \frac{\omega_1''}{\omega_0''} - \frac{\omega_2''}{\omega_0''} \right] \left[\frac{\omega_0''}{\omega} + \frac{\omega_1''}{\omega} + \frac{\omega_2''}{\omega} \right]$$

$$\times \frac{B_{\perp k'} \cdot B_{\perp p'}}{B_0^2} (p'' d)^2 \frac{A_{rp''}}{A_{rk}}$$

$$= \frac{3}{2} \sum \sum \left[\frac{k'' \cdot p'' z_{th}^2}{(\omega_0'')^2} (p'' d)^2 \right] \epsilon^0$$

$$\times \left(1 - 2 \frac{\omega_1''}{\omega_0''} + \left(\frac{\omega_1''}{\omega_0''} \right)^2 - 2 \frac{\omega_2''}{\omega_0''} + \frac{2 \omega_1'' \omega_2''}{\omega_0''} + \left(\frac{\omega_2''}{\omega_0''} \right)^2 \right) \epsilon^2$$

$$\times \left(\frac{\omega_0''}{\omega} + \frac{\omega_1''}{\omega} + \frac{\omega_2''}{\omega} \right) \frac{B_{\perp k'} \cdot B_{\perp p'}}{B_0^2} \frac{A_{rp''}}{A_{rk}} \epsilon^2$$

Find the frequency

$$\epsilon^1 \quad \frac{\omega_0}{L} = \frac{\omega^*}{L} \Rightarrow \omega_0 = \omega^*$$

$$\epsilon^2 \quad \frac{\omega_0}{L} \left(-i \alpha_{\text{eff}} \frac{\omega^{*T}}{L} \right) + \frac{\omega_1}{L} = -i \alpha_y (kd)^2$$

$$\Rightarrow \omega_1 = i \alpha_{\text{eff}} \frac{\omega^*}{L} \omega^{*T} - i \alpha_y L (kd)^2$$

$$\epsilon^3 \quad \frac{\omega_0}{L} (kd)^2 + \frac{\omega_1}{L} \left(-i \alpha_{\text{eff}} \frac{\omega^{*T}}{L} \right) + \frac{\omega_2}{L} = \frac{3}{2} \sum \sum \frac{|k'' \cdot p''| z_{\text{th}}^2}{(\omega_0'')^2} (p''d)^2 \frac{\omega_0''}{L} \frac{|B_{\perp,k} \cdot B_{\perp,p'}|}{\beta_0^2} \frac{A_{r,p'}}{A_{r,k}}$$

$$\Rightarrow \omega_2 = -\omega^* \left[\left(1 - \alpha_y \alpha_{\text{eff}} \frac{\omega^{*T}}{\omega^*} \right) (kd)^2 + \left(\alpha_{\text{eff}} \frac{\omega^{*T}}{L} \right)^2 \right] + \frac{3}{2} \sum \sum \frac{|k'' \cdot p''| z_{\text{th}}^2}{(\omega_0'')^2} (p''d)^2 \omega_0'' \frac{|B_{\perp,k'} \cdot B_{\perp,p'}|}{\beta_0^2} \frac{A_{r,p'}}{A_{r,k}}$$

$$\therefore \omega = \omega^* + i \alpha_{\text{eff}} \frac{\omega^{*T}}{L} \omega^* - i \alpha_y L (kd)^2 - \omega^* \left[\left(1 - \alpha_y \alpha_{\text{eff}} \frac{\omega^{*T}}{\omega^*} \right) (kd)^2 + \left(\alpha_{\text{eff}} \frac{\omega^{*T}}{L} \right)^2 \right] + \frac{3}{2} z_{\text{th}}^2 \sum \sum \frac{|k'' \cdot p''|}{\omega_0''} (p''d)^2 \frac{|B_{\perp,k'} \cdot B_{\perp,p'}|}{\beta_0^2} \frac{A_{r,p'}}{A_{r,k}}$$

Drake 1980: 1-25

← is this?

imaginary part

Drake 1980 (8) & 2.83.

(14)

式(24)の導出

擾動展開を線形項に留めておく

$$\omega = \omega_0 + \epsilon \omega_1$$

$$\Rightarrow \omega = \omega_* + i\alpha_{\text{eff}} \frac{\omega_*}{\omega_*} \omega_* - i\alpha_{\text{eff}} (kd)^2$$

$$\left[+ \frac{3}{2} \sum \sum \frac{|k'' \cdot p''|^2 \hbar^2}{\omega_{k''}} \frac{B_{+rk} B_{+rp'}}{B_0^2} (p''d)^2 \frac{A_{+rk}}{A_{+rk}} \right]$$

Nonlin. term を線形結果に足し合わせる。

 $\omega_{k''}$ と $\omega_{k'}$ の ω を代入

$$\begin{aligned} \omega_{k''} &= \omega_{k''} + i\alpha_{\text{eff}} \frac{\omega_{k''}}{\omega_{k''}} \omega_{k''} - i\alpha_{\text{eff}} (kd)^2 \\ &= \omega_{k''} \left(1 + i \left(\alpha_{\text{eff}} \frac{\omega_{k''}}{\omega_{k''}} - \alpha_{\text{eff}} \frac{\omega_{k''}}{\omega_{k''}} (kd)^2 \right) \right) \end{aligned}$$

$$\begin{aligned} \text{Nonlin. term} &= \frac{3}{2} \sum \sum \frac{|k'' \cdot p''|^2 \hbar^2}{\omega_{k''}} \left(1 - i \left(\alpha_{\text{eff}} \frac{\omega_{k''}}{\omega_{k''}} - \alpha_{\text{eff}} \frac{\omega_{k''}}{\omega_{k''}} (kd)^2 \right) \right) \\ &\quad \times \frac{B_{+rk} B_{+rp'}}{B_0^2} (p''d)^2 \frac{A_{+rk}}{A_{+rk}} \end{aligned}$$

第1項目は先ほどの結果と一致

(15)

α_{eff} の 2 次 2 次 整理 項 まで

$$= -\frac{3}{2} i \alpha_{\text{eff}} \sum \sum \frac{|k'' \cdot p''| \hbar^2}{\omega_{x''}} \frac{\omega_{x''}}{v} \frac{|B_{\perp k'} \cdot B_{\perp p'}|}{B_0^2} (p'd)^2 \frac{A_{r,p'}}{A_{r,k}}$$

$$= -i \frac{3}{2} \alpha_{\text{eff}} \sum \sum \frac{\hbar^2}{v} (|k'' \cdot p''|) \frac{|B_{\perp k'} \cdot B_{\perp p'}|}{B_0^2} \left(\frac{\omega_{x''} \eta_e}{\omega_{x''}} \right)$$

$$\times (p'd)^2 \frac{A_{r,p'}}{A_{r,k}}$$

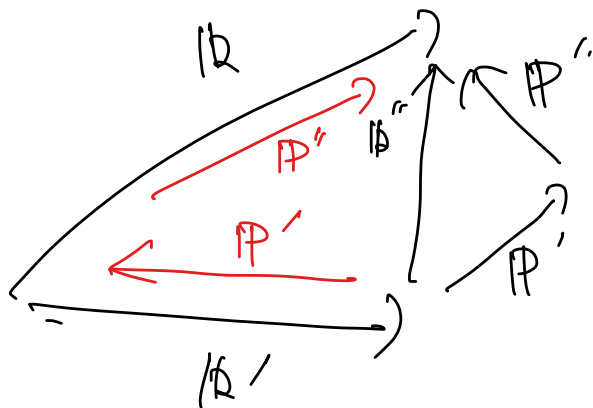
Drake の (8) の nonlin. term にも

$(p'd)^2$ は 2 次.

・ π -タリ-ク を 考慮 する 必要 あり ?

$$|k' + k'' = k, \quad p' + p'' = p'' \text{ の } S'$$

$|k|^2, \quad |B_{\perp k'}|^2$ を 考慮 する 必要 あり かも する かも



$$\left. \begin{aligned} k' &= -p' \\ p &= p'' \end{aligned} \right\} \text{ の 場合 ?}$$

$$\begin{aligned} k'' \cdot p' &= (k - k') \cdot k \\ &= |k|^2 - k' \cdot k \end{aligned}$$

$$\begin{aligned} |B_{\perp k'} \cdot B_{\perp -k'}| &= |B_{\perp k'}|^2 + \dots \end{aligned}$$